

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 21

DESCENDING ORDER

AIM: To write and execute an 8085 assembly language program for descending order and store result in memory.

ALGORITHM

1. Store the given numbers in consecutive memory locations starting from **2050H**.
2. Load the number of elements into register **C**.
3. Compare two consecutive numbers using the **CMP** instruction.
4. If the first number is smaller than the second, interchange them.
5. Continue comparing all consecutive numbers.
6. Repeat the process until all numbers are arranged in descending order.
7. Store the sorted numbers in the same memory locations.
8. Stop the program.

CODE:

```
MVI C,05H
```

```
LXI H,2050H
```

```
LOOP1: MOV A,M
```

```
    INX H
```

```
    CMP M
```

```
    JNC NEXT
```

```
    MOV B,M
```

```
    MOV M,A
```

```
    DCX H
```

```
    MOV M,B
```

```
    INX H
```

```
NEXT: DCR C
```

JNZ LOOP1

HLT

OUTPUT:

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	05
BC	05 00
DE	00 00
HL	1F 74
PSW	00 00
PC	42 1D
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

Load me at

```
1 LXI H, 8050
2 MOV C, M
3 DCR C
4 LOOP3: MOV D, C
5 LXI H, 8051
6 LOOP2: MOV A, M
7 INX H
8 CMP M
9 JNC LOOP1
10 MOV B, M
11 MOV M, A
12 DCX H
13 MOV M, B
14 INX H
15 LOOP1: DCR D
16 JNZ LOOP2
17 DCR C
18 JNZ LOOP3
19 HLT
```

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

8054 - + 02

Update Memory

Data Stack KeyPad Memory I/O Ports

Start: 8050

Address (Hex)	Address	Data
1F72	8050	5
1F73	8051	4
1F74	8052	3
1F75	8053	2
1F76	8054	1
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

00:47 11-08-2026